

UGS 303
Spring 2025



Science, Technology, and Politics

Course Information

Instructional Mode: Face-to-face

Meeting Times: Mon/Wed 02:00 PM - 03:00 PM; Fri 01:00 PM - 02:00 PM

Meeting Location: SZB 4.508; MEZ 1.216

Unique Number: 62375

Instructional Mode: Face-to-face

Meeting Times: Mon/Wed 02:00 PM - 03:00 PM; Fri 02:00 PM - 03:00 PM

Meeting Location: SZB 4.508; MEZ 1.216

Unique Number: 62380

Instructor

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TA: Raymond He

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Welcome Message

Welcome to Science, Technology, and Politics (STP)!

Whatever your major, you belong here. This is an interdisciplinary class that crosses over between social science theories and science and technology applications. You will be challenged to learn new content and to practice writing and speaking skills, using a variety of timely and

compelling materials. Please do come into this course with a "growth mindset" -- we've set the course up in a way to encourage you to push your intellectual comfort zone, learn, and improve.

Please read this syllabus carefully. Additionally, the course Canvas page is designed to lower the administrative burden on you. You can follow along for the topics, readings, and assignments through the "Modules" tab and the pages inside.

I am an Associate Professor in the Government department, and I specialize in international relations. My main field is the politics of international economic, financial, and business relations, which is also known as international political economy. You can learn more about me and my research at my webpage: <https://www.rwellhausen.com/>. Many years ago I took a class something like this STP course. It was probably one of my favorite courses ever -- I found it challenging and eye-opening. I look forward to sharing these ideas and learning with you this semester!

I am committed to creating a learning environment where diverse perspectives are recognized and valued as a source of strength. I request that all students work with me to create a class culture based on open communication, mutual respect, and inclusion. As a class we will approach all discussions with respect and civility. Disagreements and debates in academic discourse are expected and welcome, but personal attacks are never OK, and will not be tolerated. I strive to ensure an open and welcoming classroom for all students. If I ever miss the mark, please don't hesitate to come and talk to me. We are all learning together.

Best,

Prof Wellhausen.

Catalog Description

Large-group lecture and discussion class focusing on a contemporary issue. Designed to introduce undergraduates to scholarly analysis from an interdisciplinary perspective. Includes an introduction to University resources; such as research facilities; museums; and attendance at University lectures or performances as assigned. Multiple sections may be offered in the fall and spring with various topics and instructors. Restricted

Overview of the Class

Course Description: Most fights over contemporary science and technology are rooted in classic political problems of power, scarcity, and the distribution of resources. In this course, we

will learn and apply foundational social science theories to understand how government interacts with science and how scientific knowledge is incorporated (or not) into public policy. What are the limits of the government in making optimal decisions about science and technology?

Signature Course Mission: The Signature Courses at the University of Texas at Austin will connect students with distinguished faculty members in unique learning environments. By way of this rigorous intellectual experience, students will develop college-level skills in research, writing, speaking, and discussion through an approach that is both interdisciplinary and contemporary.

Teaching Modality Information: This course is in-person. In-person attendance and participation is required as part of the course grade. See "Grading Policy" below for details. For MW lectures, slides are posted to Canvas (in advance of class) and classroom recordings are made available through Lectures Online. These materials are intended for review purposes only. F discussion sections are in-person only.

Pre-Requisites for the Course

Only one of the following may be counted: Tutorial Course 302; Undergraduate Studies 302; 303.

Learning Outcomes

At the conclusion of this course, students will be able to:

- understand key social science theories and apply them to everyday circumstances.
- recognize and articulate controversies around science and technology from a public policy point of view.
- recognize and conduct positive (fact-based) analyses that are distinct from normative (opinion-based) analyses.
- better make and present arguments in both written and oral forms.

Core Courses

Institutionally Designated Option

This course fulfills three hours of the component area option of the university Core Curriculum and addresses three of the core objectives established by the Texas Higher Education Coordinating Board: communication skills, critical thinking skills, and teamwork skills.

Grading Policy

Grade scale (final grades rounded to the nearest whole number):

- 94 and above A
- 90-93 A-
- 87-89 B+
- 84-86 B
- 80-83 B-
- 77-79 C+
- 74-76 C
- 70-73 C-
- 67-69 D+
- 64-66 D
- 60-63 D-
- 59 and below F

Grade components and policies:

Summary:

- Assignments 40%
- Research project 40%
- Group presentation 10%
- Participation 10%

Details:

Assignments (40%). Assignments are typically critical thinking-based assignments, in which you will apply course content, perform social science research, reflect on course materials, and so on.

They will be completed via Canvas, whether through Canvas tools or file uploads. *Assignment content is focused on Signature Course essential elements: Critical Thinking, Information Literacy, and Writing.*

- **FAQ:** What is the deadline for assignments due on Fridays, prior to discussion sections? For fairness, assignments are due by or before the first discussion section *regardless of which section you are assigned to.*

Research project (40%). Throughout the semester, you will be working on your own research project in which you will apply social science theories from the course to an STP issue of your own choice. We will work on the project step-by-step so as to give you opportunities to develop your ideas and build to your final deliverables. This project will be individual, in the form of a solo-authored final research paper. We will spend a considerable amount of time explaining expectations for this project in the course. *Research project content is focused on Signature Course essential elements: Information Literacy, Oral Communication, Writing, Critical Thinking.*

Group presentation (10%). You and a group of colleagues working on similar topic areas will bring together your individual research projects to collaborate on a final creative, oral presentation, combining your various expertise to create an analysis beyond each of your individual work. We will spend a considerable amount of time explaining expectations for this presentation in the course. *Group presentation content is focused on Signature Course essential elements: Oral Communication, Critical Thinking.*

Participation (10%). This is an in-person course. You are expected to attend all lectures in-person. You are required to participate in-person in your assigned discussion section. Here are some FAQs about the participation portion of your grade:

- ***Presence in discussion section is not equivalent to participation.*** Challenge yourself to actively contribute; we will discuss strategies in class. Part of the Signature Course mission is to get you comfortable with oral contributions, hence the emphasis on spoken participation in section.
- ***What if I can't attend my assigned discussion section on a given day?*** Should a student be absent from discussion section, the expectation is that the student will: (1) notify the TA (in advance if possible, or as quickly thereafter as is reasonable), (2) get information and notes about missed content from a colleague (**and not from the TA), (3) complete any missed work, and (4) send the TA a Canvas message with a well-written 1-paragraph reflection on the missed content + confirmation that any missed work is completed. All steps 1-4 must be completed prior to the next discussion section. *Failure to do so = no credit for participation that day.*
 - Note that you have a commitment to your colleagues in your assigned discussion section. In an exceptional circumstance, you can attend a different discussion

section on that day (and be sure the TA is aware). If this happens more than once, however, we reserve the right to dock your participation grade.

- In the instance of repeated absence from your assigned discussion section, we reserve the right to dock your participation grade irrespective of your completion of steps 1-4.
- ***Do lectures contribute to my participation grade?*** You should NOT expect to receive a high participation grade if you are notably absent in lectures, and/or if you never participate in lecture. However, we do not take attendance in lectures, so you do not need to inform the TA/Prof of absence from lecture. You are responsible for making up material; the TA/Prof will not respond to inquiries of the style, "did I miss anything?" Our MW classroom is equipped with Lectures Online for you to review recordings after class.

Extra credit? Students will have the opportunity to take an optional, take-home final as a means of earning up to a 2% increase on the students' final class grade. No further extra credit opportunities will be provided.

Additional policies relevant to grading:

Late policies:

- Assignments submitted from 2 minutes – 24 hours past the deadline will be docked one letter grade. The assignment will be docked another letter grade for each additional 24 hours.
- If your failure to submit the assignment on time results in you being unable to complete other graded content (such as actively participating in discussion), you will not receive any special treatment and will be graded to the same standards as your colleagues.
- We reserve the right to notify you that the late policy is suspended or changed for any specific assignment.

Exceptional Circumstances:

In case of emergency that will disrupt your ability to be successful in the course: Students are expected to use Student Emergency Services (SES), <https://deanofstudents.utexas.edu/emergency/>. It is SES's job to work with you and then notify your professors on your behalf. Once notified by SES, the professor will work with you to determine the best course of action and whether additional accommodations can be made, on a case-by-case basis. Accommodations are ultimately at the professor's discretion.

More details on Student Emergency Services (SES): Student Emergency Services in the Office of the Dean of Students helps students and their families during difficult or emergency situations.

Assistance includes outreach, advocacy, intervention, support, and referrals to relevant campus and community resources. If you need to be absent from class due to a family emergency, medical or mental health concern, or academic difficulty due to crisis or an emergency situation, you can work with Student Emergency Services. SES will document your situation and notify your professors. Additional information is available at: deanofstudents.utexas.edu/emergency/ or by calling 512-471-5017.

University-approved Function + Religious Holiday Conflicts:

In general, the student must inform the professor/TA, with proper documentation, well before the deadline or scheduled exam time, so that the professor/TA can work on alternative arrangements. For religious holidays, By UT Austin policy, the student must notify the professor/TA of the pending absence at least fourteen days prior to the date of observance of a religious holy day. If the student must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, the student will be given an opportunity to complete the missed work within a reasonable time after the absence.

Academic Integrity Expectations:

Plagiarism and Misconduct: Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can include a written warning, probation, deferred suspension, or dismissal from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website.

*****See generative AI policy below.*****

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The professor/TA expect you to respect them and your colleagues enough to never ask for preferential treatment in grading outside of these parameters. In particular:

- Never ask for extra credit on demand.
- Never ask for grade changes with no intellectual basis.
- Never ask for retrospective accommodations for “exceptional circumstances” that took place before you received a grade that is not at the level you hoped. The definition of an exceptional circumstance is orthogonal to your grade.
- Do not request grade changes based on the professor’s/TA’s sympathy. But you SHOULD know that we are ready to give you ideas as to how to get closer to the grade you want.

And, many experts in knowledge accumulation, pedagogy, and higher ed are employed by UT Austin to serve you – take advantage of this perk of being here.

Generative Artificial Intelligence

The creation of artificial intelligence tools for widespread use is an exciting innovation. The University encourages all students to engage with AI responsibly and to understand that there are important limitations to using generative AI for learning.

The use of generative artificial intelligence tools (such as CoPilot or ChatGPT) in this class: is **strictly prohibited**. This includes using AI to generate ideas, outline an approach, answer questions, solve problems, or create original language. **All work in this course must be your own or created in group work, where allowed.**

If you are considering the use of generative AI but are unsure if you are allowed or the extent to which they may be utilized responsibly, please ask.

If the teaching team (professor + TA) suspects AI has been used in producing work, points will be taken off, up to and including a grade of 0 on the work. If a student would like to defend that they did not use AI, that student will make a 15 minute in-person office hour appointment with the professor within 1 week of receiving the grade. In that appointment, the student will engage in a discussion of the relevant academic content with the professor, verbally and without notes. As a result of this discussion, the professor will adjudicate whether the student demonstrated individual mastery of the content such that the grade should be increased. The professor's adjudication of the issue at that point is final, and no further appeal or discussion will be entertained.

Overview of all Major Course Requirements and Assignments

(Notes: This content is copy-pasted from pages in the course's Canvas modules. As you take the course, please rely on the modules and pages set up in Canvas, as any changes during the course of the semester will be documented in Canvas but not here. Canvas also contains more detailed information.)

***** Module: Overview of science, technology, and politics *****

Jan 13 -- Lecture: Introduction and overview

* Assignment given: [Brainstorming research interests](#)

* Assignment given: [University Lecture Series](#)

- * Assignment given: University "Gem"
- * End-of-semester assignment introduced: Individual research paper
- * End-of-semester assignment introduced: Group presentation

Jan 15 -- ** No class meeting **

Notes: TA available in the classroom to discuss course questions + for late-add students to catch up.

Jan 19 – Discussion

- * Assignment due: Brainstorming research interests

Jan 20 -- ** No class meeting (MLK Day) **

22 Jan – Lecture: Positive v. Normative analysis + Policing science

- * Assignment given: Reading reflection

24 Jan – Discussion

- * Assignment due: Reading reflection

**** Module: How do science and government interact? ****
Social science concepts: Formal and informal institutions

27 Jan – Lecture: Types of science-government interaction

- * Assignment given: Reading reflection

29 Jan – Lecture: Types of science-government interaction (cont.)

31 Jan – Discussion

- * Assignment due: Reading reflection

3 Feb – Lecture: How does science get government's attention?

- * Assignment given: Reading reflection

5 Feb – Lecture: How does science get government's attention? (cont.)

7 Feb – Discussion

* Assignment due: Reading reflection

**** Module: Why is it hard to work together to solve problems? ****

Social science theory: Collective action

10 Feb – Lecture: Collective action

12 Feb – Lecture: Collective action (cont.)

* Assignment given: Personal CA experiences

14 Feb – Discussion

17 Feb – Lecture: Collective action problems

19 Feb – Lecture: Collective action case study: Intellectual property

21 Feb – Discussion

* Assignment due: Personal CA experiences

**** Module: Social science research ****

24 Feb – Lecture: Nuts-and-bolts of social science research

* Assignment given: Assignment: Your 1 sentence reserach question

26 Feb – Lecture: Nuts-and-bolts of social science research (cont.)

28 Feb – Discussion

* Assignment due: Assignment: Your 1 sentence reserach question

* Assignment due: Pre-library session assignment

3 Mar – Librarian session (**Location TBA**)

* Assignment due: Pre-library session assignment

5 Mar – ** No class meeting (a good opportunity for you to complete the "Gem" assignment!) **

* Assignment due (by 11:59 PM): University "Gem"

7 Mar – Discussion

* Assignment given: Human Subjects training (due by 11:59 PM)

**** Module: Why is it hard to agree on what the problem is? ****

Social science theory: Precautionary principle

10 Mar – Lecture: Precautionary principle

* Assignment given: Annotated bibliography

12 Mar – Lecture: Precautionary principle (cont.)

* Assignment given: Asbestos and precautionary principle

14 Mar – ** No discussion section (happy Spring Break!) **

* Assignment due: Asbestos and precautionary principle

******* Spring Break *******

24 Mar – Lecture: Risk and uncertainty

26 Mar – Lecture: Risk and uncertainty (cont.)

28 Mar – Discussion

* Assignment due: Annotated bibliography

* Assignment given: Rough draft, research paper

**** Module: (When) can property rights help solve problems? ****

Social science theory: Coase theorem + Tragedy of the Commons

31 Mar – Lecture: Coase theorem

2 Apr – Lecture: Coase theorem (cont.)

* Assignment given: Coase theorem - exploration

4 Apr – Discussion:

* Assignment due: Coase theorem - exploration

7 Apr – Lecture: Tragedy of the Commons

9 Apr – Lecture: Tragedy of the Commons (cont.)

11 Apr – Discussion

* Assignment due: Rough draft, research paper

* In-section assignment: Rough draft peer review/discussion

**** Module: Your turn ****

14 Apr – Lecture: Giving science advice

* Assignment given: Group presentation

16 Apr – *In-class public speaking center workshop*

18 Apr – Discussion

* Assignment due: Research paper

21 Apr – Lecture: Social science research – guest speaker

23 Apr – *No formal lecture*: Use class time for group presentation prep

25 Apr – Discussion: Group presentations 1/2

* Assignment due: Group presentation

28 Apr – [in our M classroom] Group presentations 2/2

* Optional assignment given: Extra credit final

- By completing this final, you can earn up to a 2% increase on your final grade.
- To be completed online, on Canvas
- Offered during our assigned finals period, which is: Friday, May 2, 10:30 am-12:30 pm

Required Course Materials

(All available on Canvas)

Recommended Course Materials

There is no textbook for this course. Course materials for each class session are available on Canvas and organized through the "Modules" tab.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without explicit, written permission of the instructor. Unauthorized sharing of materials promotes cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

Confidentiality of Class Recordings: Any class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Getting Help with Technology: Students needing help with technology in this course should contact the [ITS Service Desk](#) or your local support unit.

Final Exam Date and Time

The final exam for this course is extra credit only, as explained in the "Grading Policy" section of the syllabus. By completing the final, a student can earn up to a 2% increase on their final grade.

The extra credit final will be offered online (via Canvas), during our assigned finals period, which is: Friday, May 2, 10:30 am-12:30 pm.

Class Canvas Site

<https://utexas.instructure.com/courses/1416994>

Notice of Academic Accommodations from Disability and Access (D&A)

In case of a disability and access issue: Students must use Disability and Access Services (D&A), <https://disability.utexas.edu/>. For accommodations to be implemented, students must have a prior accommodation letter from Disability and Access that we have discussed and signed. No accommodations will be made retroactively. Note that accommodations are ultimately determined based on the context of a specific course, as defined by the professor.

More details on Disability Accommodations: If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). You may refer to D&A's website for contact and more information: <https://disability.utexas.edu/>. Please deliver your Accommodation Letter to the professor as early as possible in the semester so we can discuss your approved accommodations and get everything sorted out sooner rather than later.

University Policies and Resources for Students Canvas Page

This Canvas [page](#) is a supplement to all UT syllabi and contains University policies and resources that you can refer to as you engage with and navigate your courses and the university.

Other Helpful Information

This is content on University Policies that we have found useful to highlight in the past. More information on these and other policies are available in the [University Policies and Resources for Students Canvas Page](#) mentioned above.

Student Health and Welfare:

Counseling and Mental Health Center (CMHC): We all benefit from support during times of struggle. Know you are not alone. If you are experiencing signs of stress, anxiety, depression, academic concerns, loneliness, difficulty sleeping, or any other concern impacting your well-being – you are strongly encouraged to connect with CMHC. The Counseling and Mental Health Center is located on campus and provides a wide variety of mental health services to all UT students, including crisis services, counseling services with immediate support and well-being resources.

Additionally, CARE Counselors are located within each of the academic schools and colleges. These counselors are familiar with the concerns that are unique to their unit's students. For more information on CMHC, visit <https://cmhc.utexas.edu/> or call 512-471-3515.

Behavioral Concerns: If you have concerns about the safety or behavior of fellow students, TAs or professors, contact BCCAL (the Behavior Concerns and COVID-19 Advice Line) at: <https://safety.utexas.edu/behavior-concerns-advice-line> or by calling 512-232-5050. Confidentiality will be maintained as much as possible, however the university may be required to release some information to appropriate parties.

University Health Services: University Health Services is the on-campus medical facility providing high quality medical care and patient education to UT students. For additional information, visit <https://healthyhorns.utexas.edu/> or call 512-471-4955.

Emergency Alerts: The Office of Campus Safety (512-471-5767) recommends that students sign up for Campus Emergency Text Alerts at: <https://safety.utexas.edu/ nbsp;nbsp;>

Title IX Reporting: Beginning January 1, 2022, Texas Education Code, Section 51.252 (formerly known as Senate Bill 212) requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence, and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with me, or with any faculty or staff member about a Title IX related incident, please remember that I will be required to report this information. If you would like to speak with a Case Manager for Support and Resources, who can provide support, resources, or academic accommodations, in the Title IX Office, please email: supportandresources@austin.utexas.edu. A Case Manager can also provide support, resources, and accommodations for pregnant, nursing, and parenting students. For more information about reporting options and resources, please visit: <https://titleix.utexas.edu>, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.